Imperial College London

AERO96005 - GROUP DESIGN PROJECT

Wholly Reusable Launch System (Recovery)

Structural Design of the Empennage and Tail Boom of a Synchropter for the Aerial Recovery of Rocket Stages

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Abstract

The concept of a synchropter poses a novel challenge for the structural design of a rocket recovery system. The contents of this report entail a brief acknowledgement of the various concepts being tried for the mission in the pre-conceptual stage, followed by the procedure that lead to the general specifications of the aircraft and its layout. Finally, the design of the tail, tail mechanism, and tail boom will be examined in more detail. Throughout the project, decisions frequently consulted existing models or literature on helicopter design, while many were also shaped from structural idealisations and first principles. A philosophy employing both topological optimisation and generative design was implemented in developing the final product.

WRR Team Members

Table 1: Team Members and Roles.

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra **Atri Sharma** Alastair Wood	Structures & CAD	WRR03
Omar Arangath Thomas Cross Zuzanna Janusz **Sam Jenner** Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai San Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
Tom Alston **Hakan Serpen** Tze Leung Wong	Flight Simulation & Testing	WRR06

Sub-team leads are shown in **bold**

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1 Introduction

This report tackles the obstacles in producing a structure that is fail-safe and robust, without hindering performance or escalating the overall cost of the mission, which was the driving factor of this project. Therefore, attention had to be delegated to manufacturing costs and processes, along with fatigue and vibrational loads, so that practical considerations were not disregarded and longevity of the recapture vehicle wasn't compromised. Cooperation with other project divisions was vital in developing a comprehensive and suitable structure for sustaining the range of loads that may be encountered in the given mission profile.

2 Pre-Conceptual Stage

2.1 Design Options

Operating from only the knowledge that one or more rocket stages had to be aerially retrieved following re-entry into the atmosphere, considerations were initially given to three different capture vehicles: fixed-wing aircraft, VTOL (vertical takeoff and landing) and the helicopter. Members from each sub-team were allocated to a different concept and collaborated to identify the trade-offs associated with each option. With multi-disciplinary insight, quantitative arguments were required to avoid subjective judgement and speculation of each case.

2.2 Investigation into VTOL Design

A VTOL configuration was considered due to generally possessing greater range and cruise capabilities over common transport helicopters, with the caveat being that they would be much faster in reaching the payload location and reacting to a missed approach. Despite having greater costs associated with manufacture and fuel efficiency, the superior range of VTOLs dismisses the need to hire a cargo ship for deployment and refuelling, thus diminishing some of the cost disparity.

The most feasible VTOL design produced was a two-engine tiltrotor, similar in magnitude to the AugustaWestland 609; with a partially excavated fuselage, similar to the Sikorsky Skycrane. The absence of an internal payload meant this configuration was optimal for reducing drag and improving hover efficiency, with the option of reeling the payload into the fuselage for stability purposes.

Using historical data to estimate the MTOW (maximum take-off weight) and power consumption was less reliable than for helicopters due to the shortage in relevant regression data. A method to establish a more reliable weight estimate involved procuring the independent component weights of VTOLs such as the Bell Boeing V-22 Osprey and the Bell XV15, as a proportion of their total weight. The fuselage ratio was reduced by a near 50% due to the removal of the cargo hold. This ratio was applied to the AW609 weight, based on the power needed to carry the rocket stages with a 50% safety margin, and extended by 20% to fit the payload. The modified skycrane configuration culminated in a 10% weight save compared to a standard AW609. With more tangible weight and power values, a cost evaluation was made by Alessandro [1] to decide on the preferred configuration.

Ultimately the conclusion was to install a helicopter for the mission, as the uncertainty in landing location would be less than 50km, marginalising the strengths of the VTOL. Overall cost and design feasibility were also major factors the helicopter succeeded the VTOL in, where further detail on the final decision can be found in the Project Coordinators' reports.

2.3 Baseline Configuration

At this stage of the design, it was necessary to determine a set of design drivers which represented a hierarchy of requirements for the vehicle. Key variables were discussed with the mission profile in mind, to obtain an optimal helicopter configuration. Decisions were justified using the highest priority design drivers: cost, manoeuvrability, technology readiness and stability. The final design comprised of intermeshing rotors, unmanned piloting, a skycrane fuselage configuration, and skids for the undercarriage. Figure 1a displays the Kaman K-max, an unmanned cargo synchropter which resembles this design the closest.

(a) Kaman K-max.

(b) Sikorsky S-64 Skycrane.

Figure 1: Recovery vehicle inspirations.

3 Mission Statement

The task of the structures team was to mediate between the different criteria set out by each sub-team with the end goal of achieving an agreeable layout which satisfies stability constraints. Before efforts were made to conduct initial sizing, it was important to consider the most important criteria in the design and the interactions between them.

Firstly, it was recognised from literature [2] the fuel tanks had the highest priority in positioning, as they would need to sit near the centre of gravity to maintain trim throughout flight. This was particularly important in the presence of a swinging dynamic load, the consequence of the dangling rocket stage, for stability and modelling reasons. Secondly, the rotor and transmission would be positioned directly above it to reduce pitching instability on adjusting rotor speed. Designing against vibrational loading proved difficult, particularly in the blades and transmission designs.

The implications of vibration and noise are minimised with the absence of a pilot, as the regulations are relaxed when these qualities don't impact the handling. But the problem remains that the sensors and systems necessary for auto-piloting could become less robust with interference from vibrations. Structurally, this presents issues with fatigue, which is mainly tackled in material selection, and potential critical failure due to resonance. The natural frequency of the final model was captured from FEA to ensure this phenomenon wouldn't occur in any feasible circumstance. Additionally, the engine and fuel tanks were isolated to prevent thermal effects from inducing creep on the airframe and posing a threat to the flammable jet fuel.

Once the individual component weights and maximum takeoff weight were established by the systems and propulsion team, the stability check could be conducted. Components were shifted around to align the overall centre of gravity with the rotor hub, all while respecting the boundary conditions detailed above. Several iterations were required, which dictated the airframe shape and the range of aerodynamic exteriors which could be selected. One outstanding concern was that the tail's weight significantly outweighed the systems and electronics stored at the front of the synchropter. This meant that some components had to be shifted fore and some aft to satisfy the constraints. It also required the tail and tail boom to have a lightweight design, which will be expanded on later in this report.

A full list of the weights and positions of components in the final design can be found in table 6 in the appendix.

3.1 Airframe Design

The internal structure holding the components together was decided to be a simple truss structure with an optimised aerodynamic shell. The arguments for using a monocoque or semi-monocoque design were higher typical strength to weight ratios but without the spatial requirements for a cabin or cockpit, a truss structure can have improved weight efficiency and allow the aerodynamics team to fit a overlay with the lowest drag and downwash properties on top. Figure 2 shows the fuselage airframe, assembled through generative design by Atri [4] and Tex [5].

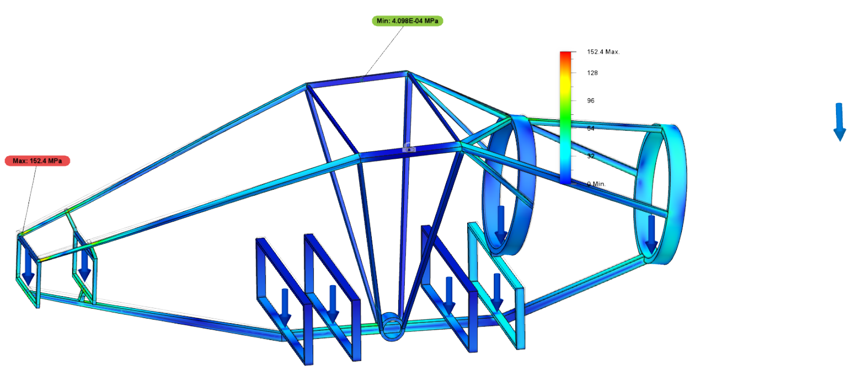


Figure 2: Finite element model of fuselage truss.

4 Preliminary Design

4.1 Material Selection

Aluminium 7075 was chosen throughout the tail structures. This was for simplicity and practicality, as composites are notoriously difficult to join to metals, and the fuselage truss was entirely aluminium. The disjointed truss structure of the fuselage frame meant it would be more expensive to manufacture and assemble in addition to the already greater material costs. Using metal permits welding which is much more robust and can offer greater strength at the joints than the parent metal does.

Opting for a composite tail boom and tail meant the joints would have to be bolted, weakening the material and increasing the susceptibility to impact loading. This would have consequences when analysing the inertial landing loads, where the joint between tail boom and fuselage would be exposed to high impact bending. Another consideration would be that the tailboom would latch onto the airframe near the engine operating at high temperatures, and composites are well known to be less creep resistant. From a fatigue perspective, aluminium is sufficient for the tail as the absence of a tail rotor means the stress would rarely peak above the fatigue limit.

4.2 Tailplane Design

4.2.1 Load Calculation

Before progressing with the tailplane structure, it was necessary to consult the FAR-29 regulations [3] to guarantee that our structure satisfied airworthiness requirements. It is required that tail loads sustained in asymmetrical yawing, and vertical drag from rotors were accounted for in addition to in-flight loads. A systematic multiplier of 1.5 was applied to all loads as a safety margin.

Kevin [6] from the aerodynamics team provided an initial geometry and airfoil to commence sizing of the internal structure, which is presented in table 2. Preliminary values, necessary for initial sizing and static stability analysis can be found under 'First iteration'. With the initial values confirmed, a more complex model was developed for the aerodynamics and flight mechanics teams to simulate and find more concrete values from. They can be found under 'Final iteration'.

	First iteration	Final iteration
Airfoil	NACA0012	NACA0012
Chord Length, c_h (m)	0.63	0.59
Span, b_h (m)	2.53	2.37
Chord Length, c_v (m)	0.68	0.62
Span, b_v (m)	1.02	0.92
Aspect Ratio, AR_h	4	4
Aspect Ratio, AR_v	1.5	1.5
Horizontal plate incidences, α (°)	-25, 9	-90, 90
Rudder incidences, α (°)	-15,15	-20,20
Tail arm, l_m (m)	4	4

Table 2: List of H-tail parameters for initial and final models.

The agreed configuration was an all-moving H-tail, or stabilator, with rudders on the vertical fins for lateral control. A greater lift capacity means greater manoeuvrability, one of the critical design drivers. It also allows for a more nimble tail which benefits both the static stability balance and in-flight stability. The short tail arm allows the design to be more compact for operation on a cargo ship, while the lower weight reflects the efforts to balance tail and systems weights. Both horizontal and vertical tails were taken as unswept and untapered

From [7], the variables which influence the incidence would be airspeed, pitch rate, lateral acceleration and collective control position. Assuming the pitch rate and control system positions remained constant throughout flight, the highest lifting on the tails occurred at max speed, with asymmetrical yawing and a tail pitched down and rudders at maximum deflection. This load case comprised of maximum negative lift, downwash and self-weight loading, defined positively downwards. These loads can be found in table 3.

Load (N)	Horizontal Tail	Vertical Tails
Max Lift, L	2700	1000
Self-weight, W	146	39
Vertical Drag, D	59	0

Table 3: Summary of in-flight loads on each tailplane.

The vertical drag over gross weight was estimated at this stage in the design using Prouty's formula in equation 1. Coefficient of drag, C_D is 2 for downwash on a plate, *(q/D.L.)* is obtained from figure 3 as a function of tailplane position along a rotor with -4° twist, A_n is the tailplane planform area and A is the rotor area. It was clear that from a structural point of view, the vertical drag on the tailplane would be significantly less impactful than the lifting load and self-weight.

$$\frac{D_V}{G.W.} = \frac{2C_{Dn} \cdot (q/D.L.)_{nn}}{A} \tag{1}$$

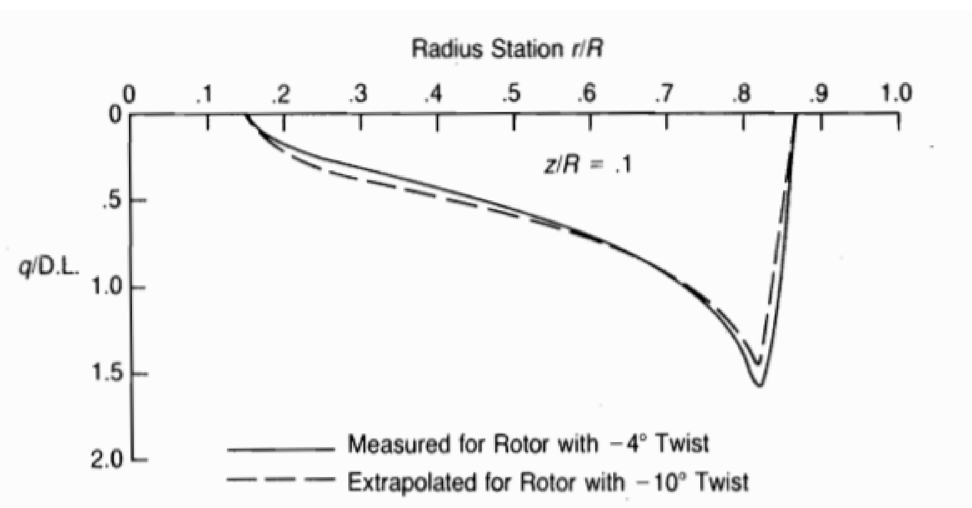
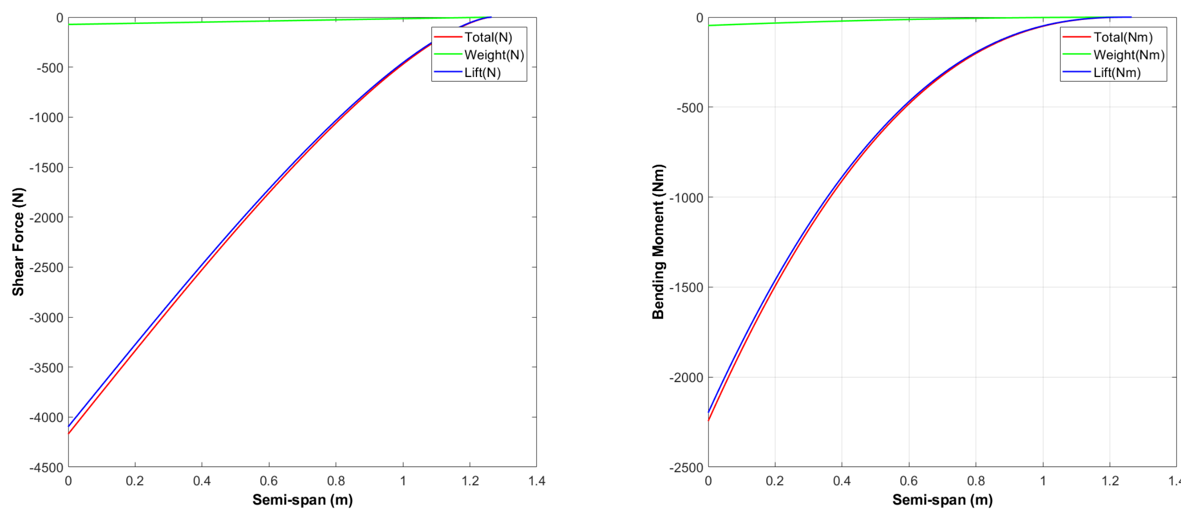


Figure 3: Dynamic pressure distribution in rotor wake.

Sizing the tailplanes relied on the load values found in table 3. The lifts were modelled as a distributed load over discrete sections along the semi-span, *b/2*, with each tail having the elliptical distribution:

$$L'(y) = L'_0 \sqrt{1 - \left(\frac{y}{b/2}\right)^2} \tag{2}$$

The L'_0 term corresponds to the entry under 'Max Lift' in table 3 for the relevant tailplane. The downwash and horizontal tail weight were modelled as linearly distributed loads along each semi-span, with vertical fin weights as point loads at the tip. The load can be integrated along the span with respect to spanwise location, y, to achieve the shear force distribution in 4a. This can be integrated again for the bending moment distribution in 4b. Every load adopted the 50% safety margin unless already done before.



(a) Shear force over span.

(b) Bending moment over span.

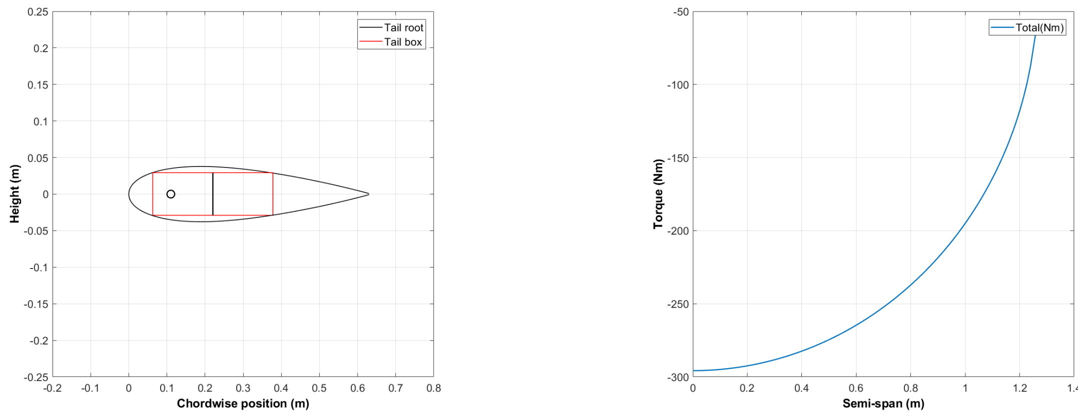
Figure 4: Shear force and bending moment distributions on the horizontal tail. Downwash is factored into lift.

With the magnitude of the bending moment, it was decided to use a two-spar wing box design. There was the option of using a single spar design to reduce potential torque and twisting of the tail but a single spar design would warrant auxilliary spars either side of it, or a very thick spar with thick skins. Each case would compound the complexity and weight of the design, with only the benefit of vacating space for the pitching mechanism.

$$T = L \cdot a + W_{tail} \cdot b - M_0 \tag{3}$$

Equation 3 calculates the torque about the flexural axis. Here, a is the distance between the A.C. and pivot, b is the distance between C.G. and pivot.

Recognising that the torque would be sizeable in an all-moving tail, it was decided to position the spars at 10% and 60% of the chord respectively, to shift the flexural axis towards the aerodynamic centre and minimise the torque needed to mobilise the tail. Figure 5 illustrates the wing box within the tail cross-section. The ring represents the A.C. position and mid-way across the box is the flexural axis.



(a) Diagram of horizontal tail cross-section.

(b) Torque over semi-span.

Figure 5: Horizontal tailplane geometry and torque distribution.

4.2.2 Spar Sizing

The sizing process begins with the spar. An I-beam idealisation is assigned to presume the spar flanges offer only bending resistance and the web offers only shear resistance. Assuming a negligible web thickness initially, the flange experiences a maximum bending stress of 116N/mm^2, far below the yielding of Aluminium 7075 T6, 572MPa. Table 4 shows the resulting spar dimensions.

Parameter	Front spar	Rear spar
Spar height (mm)	58.3	58.3
Flange width (mm)	7.3	7.3
Web thickness (mm)	1.4	1.2

Table 4: Spar dimensions

4.2.3 Stringer and Skin Sizing

(a) Typical two spar wing box.

(b) Diagram of multi-cell shear flow.

Figure 6: Tail box and shear flow idealisation [9].

Figure 6 displays illustrations of the tail cells. Equating the twist in each cell generates an equation expressing the relation between the shear flow, cell dimensions and skin/web thickness. Using vertical force equilibrium along each spanwise section, the shear force can be expressed in terms of the shear flows. Since the dimensions of the cell remain constant spanwise, the magnitude of the shear stress at each section can be directly related to the wall thicknesses, which is decided based on the shear strength of the material. Normally the thickness tapers towards the tip as shear forces diminish but in light of the extra structural load from the vertical fins, the thickness is kept constant. The thicknesses are compiled in table 5.

Parameter	Thickness (mm)
D-section	1
Spar web	0.8
Smeared panel	2.6

Table 5: Skin thicknesses

As mentioned before, it is important to minimise the weight of the tail for stability purposes. Stringers offer a weight reduction while improving the effective skin thickness and resistance to bending. Z-stringers were opted for their high strength to weight ratio.

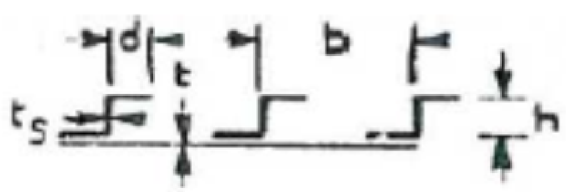


Figure 7: Z-stringer sizing conventions.

Parameter	Value (mm)
Height, h	10
Flange width, d	0.8
Spacing, b	15.2
Stringer thickness, t_s	1.3
Skin thickness, t	1.4

Figure 8: Stringer dimensions

The method for sizing the stringers was iterative. Selecting ratios of h/b and t_s/t, the ESDU sheets [10] provide constants that determine the smeared critical yield and buckling stresses. Final stringer dimensions can be found in figure 8, while figure 9 indicates the recommended number of stringers along the semi-span. In similar fashion to the skin thickness, the number of stringers will be kept constant throughout, equal to the number at the root.

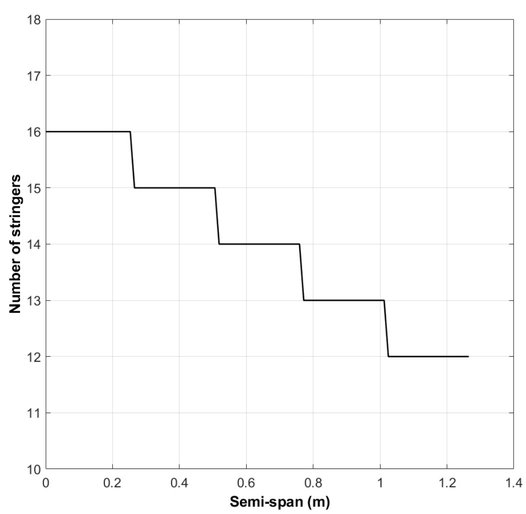


Figure 9: No. of stringers per section.

4.2.4 Rib Sizing

Ribs act as lateral stringers, to reinforce the skin against compressive buckling. The cross-section of each rib will depend on the wing box area, with the thickness being decided by the magnitude of a crushing force at each rib position. An initial guess is made to determine the number of ribs and rib spacing over the semi-span. The crushing force is determined from equation 4 to find the stress applied over each rib, so the thickness can be determined by the critical buckling stress at each point.

$$F = \frac{M^2\, L\, h_c\, t_e\, c}{2\, E\, I^2} \tag{4}$$

The critical rib thickness throughout was below 0.3 mm. Such a fine sheet would be difficult to manufacture with reasonable precision. Instead a constant rib thickness of 0.6 mm is recommended instead, with a 100% safety margin on top of an already low risk of buckling.

4.2.5 Vertical Tailplane

The vertical tails were sized in the same manner as the horizontal, with the absence of the stabilator's complex geometry and mechanism in favour of two rudders capable of deflecting twenty degrees either way.

4.3 Tail Mechanism Design

The tail mechanism required to pitch the stabilator had implications on the tail boom design, as it limited the aft cross-section and induced extra bending. A servo system was preferred over a more common hydraulics arrangement due to its greater control response, lighter architecture and greater operating efficiency. It also fulfils the condition that the tail operates with the rotor blade collective, a feature the Kamax K-max possesses for stability reasons. Servo tabs are already a common trait of helicopter tailplanes and a similar mechanism was hypothesised for the all-moving tail as the dimensions are still relatively compact.

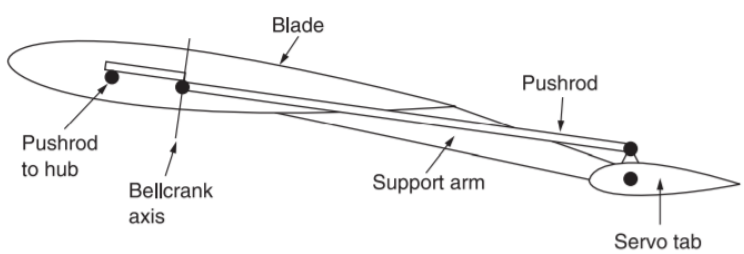


Figure 10: Typical servo tab mechanism [8].

Observing the servo tab geometry, it would be possible to implement a shaft connecting the pushrod to the trailing edge with enough torque to pivot the entire tailplane as opposed to only the control tab. The main challenges would be to provide the torque required and to maximise the torque efficiency. Torque efficiency is reliant on minimising the pushrod length by bringing the servo motor shaft closer to the trailing edge axis. This is a design case, and the geometric constraints demand either a thicker, heavier airfoil with less desirable aerodynamics is used to vacate space, or a heavier motor is used.

It is known from the prior load analysis that the peak torque experienced is approximately 290Nm. The range of stabilator incidences vary from -90° to 90°, but assuming that at high speeds the range of incidences would not exceed those installed on the UH-64, -39° to 8°, the relation in equation 5 can decide the rated power, P, of the motor, where ω is the rotational velocity, α is the incidence and t_p is the pitch rate. With a conservative pitch rate of 0.1 $\frac{deg}{deg/sec}$ for manoeuvrability, the suitable motor weight would be between 15kg to 20kg each. Given that two motors are required to control the tails separately, these values would be destabilising for the weight balance of the whole aircraft.

$$P = T \cdot \omega \tag{5}$$

$$\omega = \frac{(\Delta \cdot \alpha)\,\pi}{180\,t_p} \tag{6}$$

An optimisation process was conducted using MATLAB to minimise the weight of the mechanism. This could be achieved by employing a two stage bevel gear reduction system to amplify the torque from a smaller motor to pitch the tail. The use of bevel gears were so that the motor length ran parallel to the tail boom length rather than the shaft, enabling the lightest and most compact tail boom structure to be designed with the intent of being concealed within an aerodynamic shell. These gears operate at ratios between 3:2 and 5:1 with between 93% to 97% efficiency. Each gear ratio was tested to find the compromise between motor weight and gear weight.

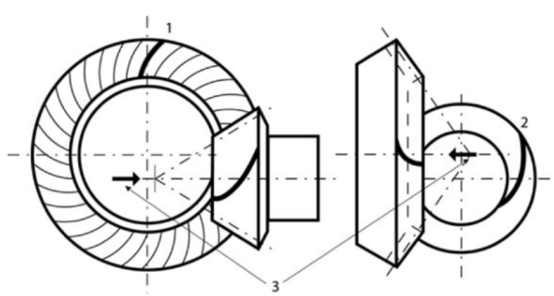


Figure 11: Spiral bevel gear setup [11].

Component	Weight (kg)
Servo motor	5.3
Gears	5.46
Gearbox	10.20
Lubricant	0.52
Total	21.48

Figure 12: Optimal motor and gear weights

The gears were sized according to a 14-tooth stub involute model, with a composition of mild steel. More properties can be found in the appendix. The gearbox was designed to be a compact aluminium casing with a 10 mm clearance from the gears so that less lubricant was needed. Main assumptions in the weight optimisation were that cooling was not needed due to the low shaft speed, and that electronics feeding to the control systems and generator were of negligible weight. Motor weights and dimensions were based on existing models. The script used to optimise the mechanism weight is attached in the appendix.

4.4 Tail Boom Design

The initial preference was for the tail boom to adopt a semi-monocoque design, with a similar design procedure to the tailplanes. This structure guarantees a high strength to weight ratio and superior torsional resistance through the skin. However, geometric constraints from the fuselage meant the engine exhaust would protrude from the bottom of the tail boom, causing a cutout in the skin and diminishing the torsional strength greatly. An open tubular frame similar to the that of the fuselage allows for heat dissipation, alleviating some material concerns, hence was utilised for this structure.

Circular rings connected by rods was determined through generative design of a model in Fusion 360. Circular cross-sections permitted the engine to extend through the tail boom length while being able to fit the aerodynamic shark skin model on top. A typical rectangular sectioned truss was considered, but rejected as the prevalent failure was through shear rather than bending.

The connecting tubes were sized according to the same method the stringers were sized in the tail design, as both are idealised to resist the full bending moment and the rods share the cross-sectional shape of the stringers to reduce weight and provide innate torsional resistance in the absence of a skin. However, this time top-hat stringers were used instead, to increase the contact area for welding, thus improving the joint strength; and for the greater compressive buckling resistance [12]. The buckling risk on the bottom rods were more concerning than the tensile on the top, as the rods had to be pushed towards the sides to facilitate the engine, thus increasing the stress near the joints and throughout the member. ESDU 98016 [14] was used for these stringers.

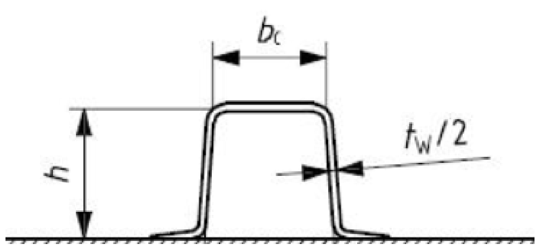


Figure 13: Omega-stringer sizing conventions.

Parameter	Value (mm)
Height, h	20
Flange width, b_c	15
Spacing at root, b	339
Stringer thickness, t_w	2
Ring thickness, t	20

Figure 14: Omega-stringer dimensions

A fail-safe design with redundancies was employed as the weight of the whole tail boom was 12.4 kg when made from Aluminium 7075-T6, less than the estimate for weight balancing. Making five members was much more efficient for resistance to torsion than three, and if a member failed the remaining structure could still function. It is also much easier to service and maintain the structure as parts could easily be welded and repaired, whereas in a monocoque structure, a birdstrike could be catastrophic for the skin's load-carrying capacity.

There were attempts to form diagonal members and X-shaped members for further torsional resistance, as commonly used in steel truss frames built for supporting concrete in buildings, but the taper of the tail boom constrained the possible geometries and prevented them from being installed. It was hypothesised that they could also lead to unwanted stress concentration at certain positions, as an X-shape is not optimal for bending.

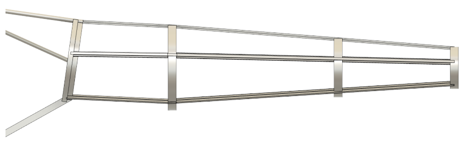


Figure 15: Tail boom truss side view.

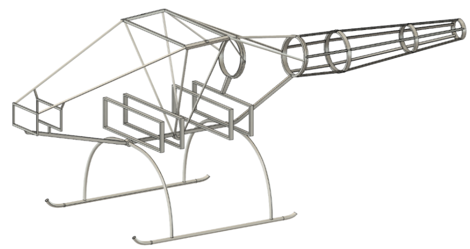


Figure 16: Full airframe model.

5 Concluding Remarks & Future Steps

Within the given timeframe, an ambitious design for towing falling rocket stages back to base was achieved. For the structures team; the uncommon rotor configuration proved to be a tedious job for Alastair [13], while the truss and unusual tail design also proved to be troublesome. These rare concepts severed the reliance on historical data and demanded that fundamental structural ideas were applied and through trial-and-error, a suitable compromise could be obtained.

The airframe design followed this philosophy, where members were added, moved and removed to optimise for strength and weight, or to investigate the fail-safe design. The mechanism design was a challenge, as with little literature on which systems would be optimal or which mechanism would be most efficient, a model had to be realised to stick to deadlines and commence with the tail boom design. With more resources, an improvement would be to conduct more research and stage a design case for multiple concepts, as was the process for selecting the airframe and structural layouts.

One more improvement would be to rigorously test for response to vibrations with CAE. As opposed to only ensuring resonance wouldn't occur, it would be more comprehensive to scrutinise how vibrations would influence each member and what kind of damping systems could be employed throughout the aircraft to tackle it.

The argument for the case of using composites could also be explored further. The material selection was unanimously biased towards metals on a cost basis and due to factors mentioned in section 4.1, but there are effective methods to combat these dilemmas like heat curing for fixing to other composites or metals, and filament winding the truss rings. It is unclear how the costs and manufacturing complexity would compare to metals but there is potential for further weight and fuel-save if composites were used.

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A Appendices

Table 6: Summary of individual component weights and locations.

Parameter	Weight (Kg)	CG Distance(m)
Fuselage Components		
Fuselage & Skids	130	0.05
Fuel Tank	55	0
Fuel Capacity	291	0
Payload Mass	1400	0
Avionics	150	-1.9
Electro Hydraulics	80	0
Fluids+Anti-Icing	62	0
Ballast	35	-2.1
Rotor		
Rotor	165.8	0
Transmission	221.7	0
Engine	268.1	1.05
Tail		
Tail Boom	12.4	2
Tail Mechanism	23	4
Tailplane	25	4
Total		
Total Weight	2922.1	0.1

B Supporting MATLAB Code

This script finds the minimum tail mechanism weight.

```
% Tail_weight_optimization_gdp.MLX
% - - - - - - - - - - - - - - - - - - - - - - - - - - -
clear
clc

Wmot=[4.6,5.8,7,8.2,9.4,3.7,4.5,5.3,6.9,8.8,10.7];%range of motor weights
w=[95,95,95,95,95,75,75,75,115,115,115];%range of motor frame sizes for sizing ...
    tail boom cross section
P=[510,900,1230,1530,1770,520,730,930,770,1530,2120];%range of rated powers (max ...
    power)
ds=[14,19,19,19,19,14,14,14,19,19,19];%shaft diameter
T=P/1.22;
Treq=2078*ones(1,length(T));%required torque at designated shaft speed
u=Treq./(0.98*0.93*T);
h=17.14;%thickness of gears, found from Lewis' equation
S=447.2;%gear steel yield stress
Np=14*ones(1,length(u));%pinion teeth
Ng=ceil(u.*Np);%gear teeth
angle=180*ones(1,length(Ng));
k=1.25;
d1=k*(((2/(0.98*0.93*S))*1000*T.*((u+1)./u))./((tand(angle./Np)))).^(1/3);%pinion ...
    diam, has to be less than 142
d=d1+ds/1.5;
D=u.*d;
t=10;
l=1.2;%factor of safety between gear and gearbox surfaces
dg=l*d;
Dg=l*D;
hg=l*h;
g=0.5*dg;
Agearbox1=(((Dg+dg+hg+(2*t))*2.*(w+g))-(2*(w+g-hg-t)).*(D+d));
Vgearbox1=Agearbox1.*(Dg+(2*t));
Vgearbox2=2*(Dg.^2+2*(dg.^2));
Wgearbox=(Vgearbox1-Vgearbox2)*2710e-9;
Wgears=7850e-9*h*2*(2*(pi*(k*d/2).^2)+(pi*(k*D/2).^2)-3*(pi*(ds/2).^2));
Wlub=(((1/3)*D).*(h+d+D)*2.*d)*880e-9;
Wtotal=Wmot+Wgearbox+Wgears+Wlub;%total weight

for i=1:1:length(T)
    if d(i)+D(i)>630 || u(i)<1.5 || u(i)>5
        T(i)=0;
        Wmot(i)=0;
        Wgears(i)=0;
        Wgearbox(i)=0;
        Wlub(i)=0;
        Wtotal(i)=0;
    end
end
%all entries of 0 in Wtotal are rejected for having too large/low gear ratio or ...
    having gears that don't fit tail boom, the next lowest value is the chosen ...
    weight
```